

VIDYA JYOTHI INSTITUTE OF TECHNOLOGY

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DEPARTMENT OF INFORMATION TECHNOLOGY

LAB PROJECT REPORT



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Project Title	:	Chat Application
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CERTIFICATE

This is to certify that the Object Oriented Programming through Java Lab project report entitled “**Chat Application** ” submitted by **Garima Choudhary(24911A1224)** to Vidya Jyothi Institute of Technology (An Autonomous Institution), Hyderabad, of B. Tech. II year, II Semester in Information Technology a Bonafide record of Lab project work carried out by them under my supervision.

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Abstract

The Chat Application is a Java-based communication system developed to enable multiple users to exchange messages in real time through a graphical user interface. Traditional communication methods may lack instant interaction and centralized messaging features; therefore, this project aims to provide a user-friendly platform for seamless communication.

The application supports multiple user messaging, a real-time chat window, and notification alerts to improve interaction between users. The project is designed using Java concepts such as GUI components, multithreading, and socket programming. Threads are used to handle simultaneous user interactions efficiently, while sockets can be utilized for communication between client and server systems.

The main objective of this project is to create a fast and interactive messaging environment where users can communicate instantly. The system provides an easy-to-use interface that allows users to send and receive messages without delay.

The expected outcome of the project is a reliable chat system that demonstrates real-time communication using object-oriented programming concepts and Java technologies.

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Chapter 1 – Introduction

1.1 Project Overview

The Chat Application is a software system developed using Java that allows users to communicate with each other in real time. The application enables multiple users to send and receive messages through a graphical interface. It provides an efficient and interactive communication platform.

Real-world applications of chat systems include customer support services, team collaboration applications, educational platforms, and social networking systems. Real-time communication has become an essential requirement in many industries

1.2 Objectives

- To develop a real-time communication platform
- To allow multiple users to exchange messages
- To provide notification alerts for incoming messages
- To reduce communication delay
- To improve user interaction and experience
- To implement Java GUI and multithreading concepts

1.3 Scope of the Project

The system is designed for users who require real-time communication through a desktop application. The project can be extended in the future by integrating cloud databases, file sharing, voice calling, and video conferencing features.

Chapter 2 – System Analysis

2.1 Functional Requirements

- User login
- Send messages
- Receive messages
- Multiple user communication
- Real-time message updates
- Notification alerts
- Exit application

2.2 Non-Functional Requirements

- Security
- Reliability
- High performance
- User friendliness
- Scalability

2.3 Software Requirements

Requirement	Details
Language	java
IDE	Eclipse
JDK	JDK 8
GUI	AWT

2.4 Hardware Requirements

Requirement	Details
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RAM	4 GB
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Processor	Intel i3
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Storage	500 GB
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Chapter 3 – System Design

3.1 Architecture Diagram

User → GUI Interface → Thread Processing → Socket Communication → Chat Server → Other Users

3.2 UML Diagrams

Mandatory:

- Use Case Diagram
- Class Diagram

Optional:

- Sequence Diagram
- Activity Diagram

3.3 Design Database

No table is required for the basic level.

TableName	Purpose
Users	Store user details
Messages	Store chat details

Chapter 4 – Implementation

4.1 Modules Description

Module	Description
Login Module	User authentication
Chat Module	Send and receive messages
Notification Module	Message alerts
Thread Module	Handle multiple users

4.2 Important Java Concepts Used

- Classes and Objects
- Inheritance
- Multithreading
- Exception Handling
- GUI using Swing
- Socket Programming
- Event Handling

Chapter 5 – Testing

5.1 Test Cases

Test Case ID	Input	Expected Output	Result
TC01	Valid Login	Login Success	Pass
TC02	Send Message	Message Delivered	Pass
TC03	Multiple Users	Simultaneous Chat	Pass
TC04	Invalid Input	Error Message	Pass

5.2 Error Handling

- IOException
- SocketException
- NumberFormatException
- NullPointerException

Chapter 6 – Results and Screenshots

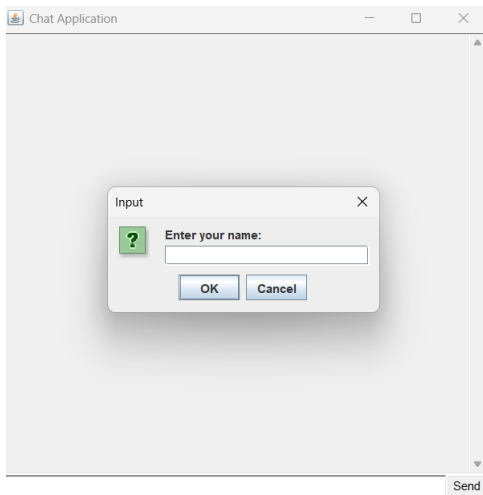


Fig 6.1 login page

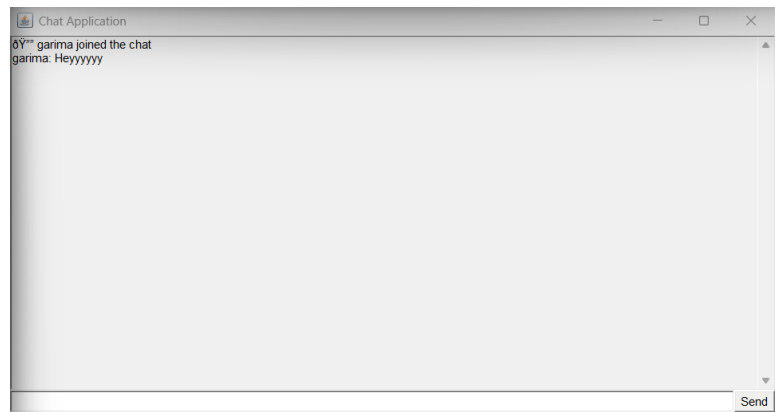


Fig 6.2 user chat application

```
C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\garim\OneDrive\Desktop\programming.2nd>javac ChatServer.java

C:\Users\garim\OneDrive\Desktop\programming.2nd>java ChatServer
Server started...
garima: heyyyyyyyyyy
garima: how is everyone Doing
garima disconnected
```

Fig 6.3 Server output screen

Chapter 7 – Conclusion and Future Scope

Conclusion

The Chat Application was successfully developed using Java technologies. The project achieved real-time communication between users and implemented GUI and multithreading concepts effectively. The project enhanced understanding of Java programming and networking concepts

Future Enhancements

- File sharing support
- Voice calling feature
- Video conferencing
- Cloud integration
- Mobile application support

References

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2. Thinking in Java — Bruce Eckel
3. Oracle Java Documentation
4. Java Swing Documentation
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